UniWeb — Extra Gaps, Recommended Sequence & Current Position Analysis

Deep Check Findings + Practical Timeline + Market Comparison
Companion document to the Priority Product Roadmap

1. Extra Gaps (Deep Check) — [redacted] Phase List [redacted]

[redacted] gaps Phase 1–3 [redacted] granular list [redacted] deep review [redacted] players (Razorpay, Cashfree, Juspay) [redacted] mature [redacted]

#	Gap / Feature	[illegible]	Priority
1	**Reconciliation Engine** Gateway settlement file vs system transactions auto match	Daily mismatch catch [redacted], finance team [redacted]	**High**
2	**Risk Engine / Velocity Rules** Same UPI high frequency, sudden spike, new merchant high volume	Fraud [redacted] chargeback [redacted]	**High**
3	**Reserve / Rolling Reserve** New merchants [redacted] 5–10% hold + release schedule	Industry standard risk cover	Medium
4	**Grievance Redressal Mechanism** Complaint ticket + escalation + SLA + RBI style policy	RBI compliance + customer trust	**High**
5	**PCI-DSS Readiness Path** [redacted] card data touch [redacted]	[redacted] UPI focus [redacted]	Later
6	**Partner Commercial Dashboard** [redacted] gateway [redacted] pricing, success rate, settlement delay	Multi-gateway manage [redacted]	Medium
7	**Merchant Health Score** KYC quality + dispute rate + volume consistency	Risk based decisions [redacted]	Medium
8	**Platform vs Partner Responsibility Matrix** Agreement + website [redacted] live acquiring [redacted]	Legal clarity + partner onboarding	**High**

Note: *[redacted] Reconciliation, Risk Engine, Grievance [redacted] Responsibility Matrix [redacted] Phase 2–3 [redacted] parallel [redacted] plan [redacted]*

2. Recommended Sequence (Practical Timeline)

Week 1–2

Focus: Security fix + Payout foundation

- Security & Secrets Hardening [redacted] (password change, .env, credentials cleanup)
- Payout partner final [redacted] (Cashfree Payout / RazorpayX / Decentro)
- Sandbox credentials [redacted] basic Create Payout + Status API integrate [redacted]
- Failed Reason Mapping dictionary [redacted] UI [redacted]

Week 3–4

Focus: Real money flow usable [redacted]

- Webhook receiver + signature verification + auto status update

- Beneficiary Management + Penny Drop
- Bulk Payout (CSV upload)
- Maker-Checker for high value payouts
- Staff MFA mandatory

Month 2

Focus: Scale + Multi-gateway readiness

- Smart Routing + Fallback
- Multi-Gateway Status Matrix
- One-click KYC Forward to selected gateways
- High-throughput Dynamic QR + Multiple Virtual Accounts
- Webhook Reliability (Idempotency + Retry + Dead Letter)

Month 3 onwards

Focus: Professional + Compliance layer

- DigiLocker + Auto KYC verification
- eSign for Merchant Agreement
- Immutable Audit Trail
- Dispute / Chargeback full workflow
- Advanced Reporting + Risk Engine + Reconciliation
- Grievance Redressal + Rolling Reserve
- Sub-merchant model + White-label + Recurring (as needed)

3. Current Position Analysis

[redacted] platform **Decentro / early-stage Cashfree** [redacted] solid foundation [redacted] Admin portal, Merchant portal, Nodal separation concept, MFA, Electronic agreement, Settlement workflow, KYC upload structure — [redacted]

Market Comparison Snapshot

Area	UniWeb (Now)	Decentro / Early Cashfree	Razorpay / Juspay Level
Payout	Manual UTR entry	API based payout	Full payout + bulk + beneficiary + auto status
Routing	Single / basic	Multi partner support	Smart routing + success-rate fallback
KYC	Upload + Video + Manual review	API verification available	DigiLocker + Auto verify + Risk scoring
Settlement	Wallet → Pending → Manual complete	Scheduled + API	T+0/T+1 engine + auto + reserves
Risk	Basic controls	Some velocity rules	Full risk engine + health score
Reconciliation	Manual / limited	Partner reports	Auto match + mismatch resolution
Audit / Compliance	Some logs + e-agreement	Stronger audit	Immutable trail + grievance + FIU ready
Scale (QR / TPS)	Basic QR	Good volume support	High-throughput QR + multi VA

Biggest Gaps vs Top Players

Razorpay / Juspay [redacted] scale [redacted] automation [redacted] missing [redacted]:

- **Payout** — Manual UTR [redacted] Automatic API payout [redacted] gap [redacted]
- **Routing** — Smart multi-gateway routing + fallback [redacted]
- **Risk + Reconciliation** — Full risk engine [redacted] auto reconciliation missing [redacted]

[redacted] ***Phase 1 ([redacted] Security + Payout API)*** [redacted], [redacted] *platform real money* [redacted] *Phase 2 scale* [redacted] *Phase 3 professional + compliance layer* [redacted]

4. Quick Action Summary

Timeframe	What to Finish	Outcome
Week 1–2	Security hardening + Payout partner + Sandbox integration + Reason mapping	Safe foundation + payout path ready
Week 3–4	Webhook auto status + Beneficiary + Bulk + Maker-Checker + Staff MFA	Real money can flow end-to-end
Month 2	Smart routing + Status matrix + KYC forward + High TPS QR + Webhook reliability	Multi-gateway + higher volume ready
Month 3–5	Auto KYC + eSign + Audit + Dispute + Risk + Reconciliation + Grievance	Near Razorpay/Cashfree professional level

Document Control

Version: 1.0 | Date: August 2026 | Companion to: UniWeb_Priority_Roadmap.pdf
Owner: Uniweb Technologist PVT LTD | For internal planning only

UniWeb — Final Readiness Checklist

Goal: PA License [redacted] product/tech/ops [redacted]
Decentro · Cashfree · Razorpay · PayU · Juspay · Worldline · Banks [redacted]
Complete work list — [redacted] | High-Throughput QR included

[redacted]: [redacted] advisor / partner / merchant [redacted] — “Technical [redacted] operations [redacted] complete [redacted] structural [redacted]: [redacted] RBI Payment Aggregator License[redacted]”

1. PA License [redacted] ([redacted])

[redacted]	[redacted] PA License	[redacted] cover [redacted]
[redacted] Escrow / Nodal [redacted] customer money	[redacted]	Partner (Cashfree/Razorpay/Bank) escrow use [redacted]
[redacted] Payment Aggregator [redacted]	[redacted]	“Technology / Platform partner” [redacted]
Direct merchant acquiring [redacted] licensed partner	[redacted]	[redacted] live payments licensed partner [redacted]
RBI [redacted] PA [redacted] report	[redacted]	Partner compliance + [redacted] due diligence
Full white-label “[redacted] gateway” claim	Limited	Brand UI [redacted]; acquiring partner [redacted]

2. Must-Have Checklist (License [redacted] — [redacted])

2.1 Security & Secrets

- [redacted] passwords / API keys .env [redacted], web root [redacted]
- Production DB password [redacted]
- Code / ZIP / repo [redacted] real credential [redacted]
- Merchant + Staff login [redacted] rate limiting + lockout
- Strong password policy + secure session cookies
- Admin + Staff MFA mandatory
- KYC uploads folder protected + access logging

2.2 Real Money Flow — Payout & Settlement

- Payout API integrated (Cashfree Payout / RazorpayX / Decentro [redacted])
- Merchant Settle Now → automatic bank transfer (IMPS/UPI/NEFT)
- Webhook [redacted] auto status + UTR (manual UTR entry [redacted])
- Failed payout → reason save + auto refund to merchant wallet
- Beneficiary add + Penny Drop verification
- [redacted] Verified beneficiary [redacted] payout allow
- Bulk Payout (CSV/Excel upload + preview + process)
- Maker-Checker for high value payouts (threshold configurable)
- Settlement schedule (T+0 / T+1 / custom) + bank holiday calendar
- Wallet breakdown: Available / In-transit / Hold / Reserve

2.3 Multi-Gateway & Routing

- Multiple partners register (Razorpay, Cashfree, PayU, Decentro, etc.)
- Smart routing rules (success rate, cost, volume, category)
- Primary + Fallback gateway; auto failover on failure/timeout
- Per-merchant [redacted] global routing preference (Admin control)
- Multi-gateway status matrix per merchant (Pending/Approved/Live/Rejected)
- One-click Forward KYC docs/status to selected gateways (secure, no raw email)
- Partner commercial dashboard: pricing, success rate, settlement delay

2.4 High-Throughput QR + Virtual Accounts (Dialer / High Volume)

- Dynamic QR generation optimized for high TPS (queue + cache where needed)
- Same QR [redacted] multiple rapid payments handle (no single-use bottleneck unless required)
- QR amount lock + expiry controls
- Multiple Virtual Accounts per merchant (Store-1, Store-2, Campaign, Online…)
- Collection VA [redacted] Payout VA [redacted] mapping
- Rate limit + back-pressure [redacted] partner API limit breach [redacted]
- High-volume test: burst traffic (e.g. hundreds of QR payments in short window) sandbox [redacted]
- Clear merchant UI: which VA / QR is for which use-case
- Reconciliation support for high volume (batch match by UTR / VA / time window)
- Alert when QR/VA success rate drops or partner throttling detected

2.5 Risk Engine

- Velocity rules (same UPI / phone / IP / device in time window)
- Amount spike vs merchant average
- New merchant volume hold (first 7–15 days)
- Blacklist / Whitelist (UPI, phone, IP, device, card BIN)
- Risk score 0–100 + action bands (Allow / Flag / Hold / Block)
- Hold = capture but settlement blocked until release
- Admin rule config UI (thresholds without code deploy)
- Manual override + reason + audit log
- Merchant health / risk score rollup

2.6 Reliability & Webhooks

- ■ Webhook signature verification on every event
- ■ Idempotency keys — duplicate webhook process ■ ■■
- ■ Retry queue + Dead Letter queue + Admin reprocess
- ■ Failed reason mapping (technical → simple Hindi + English)
- ■ Merchant dashboard + email/notification ■■■ mapped reason

2.7 KYC / Onboarding / Agreement

- ■ Entity KYC + Video KYC flow
- ■ DigiLocker / auto-verify assist (PAN, Aadhaar, GST, Udyam, Bank) ■■■■ partner API ■■■■
- ■ eSign for Merchant Agreement (versioned)
- ■ Electronic acceptance record (IP, time, fingerprint) already strong — maintain
- ■ KYC status per gateway matrix

2.8 Ledger, Nodal Concept & Money Separation

- ■ Merchant wallet ledger (liability) vs Platform commission wallet ■■■
- ■ Nodal/escrow account records + “separate from platform payout bank” checklist
- ■ Real customer funds partner escrow ■■■; platform ■■■■■ instruction + ledger
- ■ Split settlement: commission cut at txn level, clear Gross → Commission → Net

2.9 Compliance Feel (■■■■ PA License ■■ ■■)

- ■ Immutable audit trail (login, KYC, payout, settlement, role, settings)
- ■ Grievance / complaint ticket (customer without login) + ticket ID + SLA + escalation
- ■ RBI-style grievance policy page on website
- ■ Dispute / chargeback workflow + evidence upload + deadlines
- ■ Rolling reserve option on new / high-risk merchants
- ■ Reconciliation: gateway file vs system txns auto-match + mismatch report

2.10 Legal Positioning & Website Trust

- ■ Website + Agreement: “UniWeb is a technology platform; live acquiring by RBI-authorised partners”
- ■ ■■■ “We are a Payment Aggregator” claim ■ ■■■■
- ■ Trust / Compliance / Responsibility Matrix page (money ■■■■, settlement ■■■, dispute ■■■■■)
- ■ Settlement timelines: “depend on partner bank/PA” ■■■ ■■■■ ■■
- ■ Active partners list (only after approved) optional on Trust page
- ■ Terms, Privacy, Refund, Merchant Agreement, System Status pages live ■■ accurate

2.11 Reporting, Support & Ops Polish

- ■ Financial + Settlement + Success-rate reports (by gateway / method)
- ■ Tax-ready / GST friendly exports
- ■ Merchant monthly statement style export
- ■ Support SLA documented (response time expectations)
- ■ Status page (platform + partner gateway health)
- ■ Incident communication template (■■ gateway down ■■)

3. High-Throughput QR — Detailed Spec (Dialer / High Volume Use Case)

■■■■ use-case ■■■ ■■ QR / VA ■■ ■■■■ ■■■■ ■■ payments ■ ■■■■ ■■■ (dialer / high-repeat small-ticket). ■■■■ ■■■ ■■■ ■■ design ■■■■■ ■■■

3.1 Technical Requirements

- QR create API fast path — minimal DB locks; pre-generate or cache static params where safe
- Idempotent payment notification handling (same UTR / same partner txn id twice = once)
- Async webhook processing (queue) so burst traffic UI/API ■■ block ■ ■■■
- Partner-wise rate limit awareness — circuit breaker ■■■ partner 429/5xx ■■
- Multiple Dynamic QRs ■■ multiple VAs per merchant ■■■■ load ■■■ ■■■
- Optional: amount-fixed QR for dialer scripts (less open amount abuse)
- Monitoring: payments/minute per QR, per VA, per merchant; alert on drop or spike anomaly

3.2 Product / Merchant Side

- Merchant portal: “High volume QR” create option with recommended settings
- Label QRs (Campaign / Counter-1 / Agent-Team-A)
- Live counter on dashboard (today’s count + amount on this QR)
- Auto-disable QR on abuse signal (risk engine link)
- Export of all payments against one QR for reconciliation

3.3 Partner Conversation Points (Decentro / Cashfree ■■■)

- Ask for high-throughput / high TPS QR or Collect limits in writing
- Ask for multiple VAs per merchant limits
- Share realistic peak: payments per minute per QR (without over-claiming)
- Confirm webhook delivery guarantees + retry behaviour under load
- Confirm settlement and reconciliation file frequency for high volume

4. Responsibility Matrix (Website + Agreement ■■■ ■■■■ ■■■■■)

Topic	UniWeb (Platform)	Licensed Partner / Bank
Customer funds custody	Ledger only — funds not in platform operating account	Escrow / nodal as per partner licence
Payment acquiring	Technical routing + merchant experience	Actual acquiring & settlement rails
Payout to merchant bank	Instruction + beneficiary verify + audit	Bank transfer execution (IMPS/UPI/NEFT)
KYC ultimate acceptance for live rails	Collect, review, forward, monitor	Partner may re-check / approve for their rails
Dispute / chargeback final	Evidence collection + merchant coordination	Network / issuer / partner rules decide
First-line support	UniWeb support + ticket system	Escalation for partner-side failures

5. Strong-to-Have (Professional Level)

- Sub-merchant / marketplace hierarchy (optional business model)
- White-label checkout (logo, colours, thank-you page)
- Recurring / UPI Autopay subscriptions
- Advanced analytics (cohort, gateway comparison graphs)
- Mobile-responsive polish on every portal (already partially done — keep tight)
- Staff granular permissions beyond basic roles
- PCI path documentation only if card data ever touches your servers

6. Recommended Order (Practical)

When	Finish	Outcome
Week 1–2	Security + Payout API sandbox + Reason mapping + Legal wording draft	Safe foundation
Week 3–4	Live payout webhook + Beneficiary + Bulk + Maker-checker + Staff MFA	Real money end-to-end
Month 2	Routing + Status matrix + High-TPS QR/VA + Webhook reliability + Risk basic	Scale + multi-partner
Month 3–5	Full Risk score + Reconciliation + Grievance + Audit + eSign/DigiLocker assist + Reserve	Only License feels missing

7. Final Position Statement (Use when ready)

“UniWeb is a technology platform for merchant onboarding, collection UX, routing, payout instructions, risk controls and reporting. Live payment acquiring and fund settlement are performed by RBI-authorised partners. Our product and operations stack is built to professional standards; the remaining structural upgrade path is obtaining our own Payment Aggregator authorisation when capital and regulatory readiness allow.”

■■ ■■■ Must-Have list ■■■■ ■■■■ ■■ — ■■ Decentro / Cashfree / Razorpay / PayU / Juspay / Worldline / Banks ■■ ■■■■■ ■■■■■■■ ■■■■ ■■■ **PA License ■■■■■ ■■■■ ■■■■■■ ■■■■** gaps product/ops ■■ ■■■■, regulatory category ■■ ■■■■■■

Document Control

Version: 1.0 | August 2026 | Uniweb Technologist PVT LTD
Use with: Priority Roadmap, Extra Gaps, Risk Engine PDFs — this is the master “only license gap” list.

UniWeb — High-Throughput QR Architecture

Flow + Developer Tasks + Partner Questions

Dialer / High-Volume UPI Collection | PA License [redacted] — Partner rails + [redacted] architecture

***[redacted]:** High-Throughput QR **PA License** [redacted] Acquiring licensed partner (Cashfree / Razorpay / Decentro / Bank) [redacted] through [redacted] TPS limit partner [redacted] commercial + risk approval [redacted] UniWeb [redacted] system burst traffic [redacted] — [redacted]*

1. Architecture Flow

```
Merchant / Dialer App
  ■
  ▼
UniWeb API — Create Dynamic QR / Assign VA  (fast path, minimal lock, cache where safe)
  ■
  ▼
Partner API (Cashfree / Razorpay / Decentro / Bank Collect)
  ■
  ▼
Customer pays via UPI  ← many payments per minute on same QR or VA
  ■
  ▼
Partner Webhook → UniWeb Queue (async, idempotent by UTR / partner_txn_id)
  ■
  ■■ Ledger credit (Available / In-transit rules)
  ■■ Risk Engine check (velocity on this QR/VA)
  ■■ Merchant live counter + optional agent dashboard
  ■
  ▼
Settlement instruction → Partner Payout (when merchant settles)
```

Component Responsibilities

Component	Responsibility
UniWeb QR/VA Service	Fast create, labels, multi-QR/VA per merchant, expiry/amount lock
UniWeb Webhook Worker	Queue consume, idempotency, ledger write, risk hook, retries
Risk Engine	Per-QR / per-VA velocity; abnormal spike → flag or temp disable QR
Partner Rails	Actual UPI collect, TPS quota, escrow, settlement file
Merchant Portal	Create high-volume QR, live counters, export by QR/VA
Admin / Ops	Partner throttle alerts, success-rate drop, reprocess dead letter

2. Developer Task List (Granular)

Phase A — Foundation (1–2 weeks)

- API: create dynamic QR with amount lock + expiry + merchant label
- API: create / list multiple Virtual Accounts per merchant (purpose tags)
- Store partner QR id / VA id mapped to uniweb_qr_id / uniweb_va_id
- Webhook endpoint: verify signature, push to queue (do not process inline under load)
- Idempotency table: unique(partner_txn_id) or unique(utr + merchant_id)
- Worker: credit ledger once; ignore duplicate webhooks safely
- Basic live counter: payments count + amount today per QR (cache Redis optional)

Phase B — Throughput & Safety (next 2 weeks)

- Queue + concurrency limits; dead-letter on repeated failure
- Circuit breaker when partner returns 429 / sustained 5xx
- Rate-limit awareness: configurable max create-QR / sec per merchant
- Risk: velocity on same QR id / same VA (e.g. N payments in M seconds → flag)
- Admin alert: success rate drop or partner throttling detected
- Merchant UI: High-volume QR create wizard + recommended settings
- Export all txns for one QR / one VA (CSV) for reconciliation

Phase C — Scale Polish

- Pre-warm / cache static QR assets where partner allows static+callback model
- Horizontal worker scale (multiple consumers on same queue)
- Dashboard graphs: payments/minute per QR, per merchant
- Auto temp-disable QR on severe risk signal (with admin re-enable)
- Batch reconciliation job keyed by VA + time window + UTR list
- Load test in sandbox: burst (document actual numbers you achieve)

3. Suggested Data Fields

```
qr_codes: id, merchant_id, partner, partner_qr_id, amount_lock, currency, expiry_at, label, mode (high_volume/normal),
status, created_at
virtual_accounts: id, merchant_id, partner, partner_va_id, purpose, status, created_at
webhook_idempotency: id, partner, partner_txn_id, utr, txn_id, processed_at
qr_stats_daily: qr_id, date, payment_count, amount_sum, success_count, fail_count
```

4. Partner Questions (Cashfree / Razorpay / Decentro / Bank)

■■ ■■■■ email / call ■■■ ■■■■■ Over-claim ■■ ■■■■ — realistic peak ■■■■■■

Technical Limits

■ Dynamic QR / Collect API ■■ ■■■ documented TPS ■■ recommended max payments per minute per QR?
■ ■■ merchant ■■ ■■■■■ simultaneous Dynamic QRs allow ■■■?
■ ■■ merchant ■■ ■■■■■ Virtual Accounts allow ■■■? Increase process ■■■■ ■■?
■ Webhook delivery: at-least-once? Retry schedule? Signature scheme?
■ Under load ■■■ webhook delayed ■■ ■■ payment status API ■■ reconcile ■■ ■■■■ ■■■ — SLA?
■ Idempotency key support create-order / create-QR ■■ ■■ ■■■■?

Commercial & Risk

■ High-volume / high-repeat small-ticket (e.g. dialer style) use-case approve ■■■■ ■■ ■■■■?
■ Peak volume ■■ ■■■ ■■■ pricing ■■ security deposit / reserve ■■■■ ■■?
■ New merchant ■■ initial daily volume cap ■■■■ ■■■■ ■■? Increase ■■■■ request ■■■■?
■ Settlement cycle high volume ■■ ■■ same (T+0/T+1) ■■■■■ ■■ ■■■?

Reconciliation & Support

■ Settlement / transaction report file frequency (real-time API vs daily file)?
■ High volume ■■■ ■■ support escalation path ■■■ ■■ ■■?
■ Sandbox ■■■ burst testing allow ■■ — limits ■■■■ ■■■?

5. Sample Wording for Partner Email

We are building a B2B collection platform (UniWeb). Merchants include shops and service providers who need Dynamic QR and Virtual Accounts. Some use-cases involve higher frequency of smaller UPI payments on the same QR/VA (e.g. counter or agent-assisted collections). We do not require to hold customer funds ourselves; we will use your Collect + Payout rails. Please share: (1) recommended TPS / practical limits for Dynamic QR and multiple VAs per merchant, (2) webhook retry behaviour, (3) process to raise volume limits after go-live, (4) sandbox options to validate burst handling. We will implement idempotent webhook processing and monitoring on our side.

6. ■■■■ ■■■■■ ■■■■ (License confusion clear)

■■■■	■■■■■
■■■ ■■ PA License ■■■■■ high TPS QR ■■ ■■■	■■■■ — partner rails ■■■■
Customer money ■■■■ current account ■■■ ■■■■	■■■■ — partner escrow
■■■ UPI switch ■■ ■■■■■■	■■■■ — partner ■■ through

7. Success Criteria (■■ ■■■■ ■■■ ready)

■ Sandbox ■■■ burst test documented (e.g. N payments in M minutes on one QR without double-credit)
■ Duplicate webhooks ■■ double ledger credit ■■■ ■ ■■
■ Merchant portal ■■ per-QR live count ■■■ ■■■■
■ Partner throttle / failure ■■ alert + circuit breaker ■■■ ■■■
■ Admin ■■ QR ■■ risk ■■ temporarily disable ■■ ■■■
■ Partner ■■ volume limit increase ■■ written path clear ■■

■■■■■■: *High-Throughput QR ■■ ■■■ ■■■■ License ■■ ■■ ■■■■ ■■■■■■ Bottleneck = Partner TPS/risk limit + ■■■■ queue/idempotency/monitoring quality■*

Document Control

Version: 1.0 | August 2026 | Uniweb Technologist PVT LTD
Companion to: Final Readiness Checklist + Risk Engine + Priority Roadmap PDFs

UniWeb Priority Product Roadmap

Complete Development Roadmap — Phase 1 to Phase 3
Real Money Flow → Scale → Professional Compliance

Phase	Focus	Duration	Outcome
Phase 1	Foundation to Live	3–4 Weeks	Real money + Security
Phase 2	Scale + Multi-Gateway	Month 2–3	High volume ready
Phase 3	Professional + Compliance	Month 3–5	Razorpay/Cashfree level

PHASE 1 — Foundation to Live (3–4 Weeks)

Goal: Real money flow + Basic scale + Security hardening. Platform becomes capable of handling live transactions safely.

1. Security & Secrets Hardening (P0)

Current State: Admin MFA exists, CSRF on some forms, basic login attempt limiting, KYC folder protected.

Tasks:

- ■ Move all passwords, DB credentials, API keys to .env file
- ■ Remove real credentials from db.local.php and similar files
- ■ Keep .env outside web root + add to .gitignore
- ■ Change production database password immediately
- ■ Remove credentials from SFTP/deployment scripts
- ■ Review file permissions on uploads/kyc and sensitive folders
- ■ Disable PHP error display in production
- ■ Add rate limiting + lockout on Merchant login
- ■ Enforce strong password policy
- ■ Set secure session cookies (HttpOnly, Secure, SameSite)
- ■ Improve Admin + Staff activity logging

***Done when:** No credentials in code/public files + brute-force protection on both portals.*

2. Real Payout API Integration (P0)

Current State: Manual UTR entry for settlements. Wallet → Pending → Complete/Fail flow exists.

Tasks:

- ■ Select payout partner (Cashfree Payout / RazorpayX / Decentro)
- ■ Obtain sandbox credentials and complete production KYC
- ■ Integrate Create Payout API
- ■ Integrate Payout Status Check API
- ■ Build secure webhook endpoint (success / failed / reversed)
- ■ Implement webhook signature verification
- ■ Add Idempotency key support
- ■ Add 'Send Payout' button on pending settlements
- ■ Auto-update status to Processing → Completed with UTR
- ■ On failure: save reason + auto-refund to merchant wallet
- ■ Test success + failure paths in sandbox

- Test duplicate webhook handling

Done when: *Merchant clicks Settle Now → money goes to bank automatically → UTR appears.*

3. Automatic Settlement Status via Webhook (P0)

- Secure webhook receiver endpoint
- Signature verification on every webhook
- Map partner statuses (Success / Failed / Reversed)
- Update settlement table automatically
- Refund wallet on failure
- Logging + alert on webhook processing failure
- Retry mechanism where supported by partner

4. Failed Reason Mapping (P0)

- Build dictionary of common gateway error codes
- Map technical codes to simple Hindi + English messages
- Show mapped reason on Transaction detail page
- Show mapped reason on Merchant dashboard
- Include mapped reason in email/notifications
- Fallback message for unknown errors

Example: ERR_INSUFF_FUNDS → 'Customer account has insufficient balance'

5. Beneficiary Management + Penny Drop (P1)

- Beneficiary add form (Name, Account, IFSC, UPI)
- Integrate Penny Drop / Account Verification API
- Name match check against bank response
- Status: Pending / Verified / Failed
- Allow payout only to Verified beneficiaries
- Show verification status to merchant
- Re-verify option

6. Bulk Payout — CSV / Excel (P1)

- CSV template download (Account, IFSC, Amount, Reference)
- Upload + validation (format, IFSC, amount limits)
- Preview screen showing success/error rows
- Process via background queue
- Individual status tracking
- Final summary report
- Support re-upload of failed rows

7. Maker-Checker for High Value Payouts (P1)

- Set configurable threshold (e.g. ■50,000 or ■1,00,000)
- Above threshold → status = Pending Approval
- UI for second Admin/Staff to Approve / Reject
- Reject reason mandatory
- Audit log for both actions
- Notification (email/WhatsApp) to approver

8. Nodal Account Real Balance Visibility (P1)

- Mark primary nodal account clearly
- Show Ledger balance vs Actual bank balance (where API available)
- Alert on mismatch
- Strengthen separation checklist UI
- Keep Platform commission wallet fully separate

9. MFA Expansion (P1)

- Make MFA mandatory on Staff login
- Give Merchants option to enable MFA
- Generate + download backup codes
- Extra verification when disabling MFA
- Admin option to force MFA reset

Phase 1 Suggested Order: Security → Payout API + Webhook → Failed Reason Mapping → Beneficiary → Maker-Checker → Bulk Payout → MFA + Nodal

PHASE 2 — Scale + Multi-Gateway (Month 2–3)

Goal: Handle high volume, multiple payment partners, and reliable operations. Platform becomes multi-gateway capable.

1. Smart Routing + Fallback (P1)

- Register multiple gateways (Razorpay, Cashfree, PayU, Decentro etc.)
- Build routing rules (Success rate, Cost, Volume, Merchant category)
- Set Primary + Fallback gateway
- Track real-time success rates
- Auto failover on primary failure/timeout
- Max retry limit
- Log every routing decision
- Admin panel to toggle routing rules
- Per-merchant or global routing preference
- Manual force-gateway option for testing

Done when: *One gateway down does not break payment flow.*

2. Multi-Gateway Status Matrix (P1)

- Show gateway-wise status per merchant (Pending / Submitted / Approved / Rejected / Live)
- Separate columns for Razorpay, Cashfree, PayU, Decentro etc.
- Last updated time + remarks
- Filter + Search
- Export option

3. One-click Forward KYC to Gateways (P1)

- Select merchant KYC documents
- Multi-select gateways
- One-click 'Forward to Selected Gateways'
- Create submission record (who, when, which docs)
- Auto status update where partner API allows
- Notify merchant
- Never email raw KYC documents — use secure method or partner link

4. High-Throughput Dynamic QR + Multiple Virtual Accounts (P1)

- Optimize Dynamic QR generation for high volume
- QR expiry + amount lock
- Logic to handle multiple payments on same QR
- Create multiple Virtual Accounts per merchant
- Map VAs by purpose (Store-1, Store-2, Online etc.)
- Keep Collection VA and Payout VA separate
- Cache / optimize QR generation API
- Add rate limit and queue where needed

5. Real-time Split Settlement (P2)

- Calculate platform commission at transaction time
- Post Merchant share + Platform share to separate ledgers
- Admin-configurable split rules (percentage / fixed)

- Allow settlement only after split
- Clear reporting: Gross → Commission → Net

6. Advanced Ledger System (P2)

- Break wallet into: Available / In-transit / On Hold / Reserve
- Ledger entry for every movement
- Clear visibility for both Merchant and Admin
- Manual adjustment with reason + audit

7. Settlement Schedule Engine (P2)

- Support T+0 / T+1 / T+2 / Custom schedules
- Bank holiday calendar
- Merchant-wise schedule override
- Auto settlement cron job
- Manual 'Settle Now' continues to work alongside schedule

8. Webhook Reliability — Idempotency + Retry + Dead Letter (P2)

- Unique event ID on every webhook
- Idempotency key to prevent duplicate processing
- Failed webhooks go to retry queue
- After max retries → Dead Letter queue
- Admin view of dead letter events + reprocess option
- Proper logging

9. Rate Limiting & Brute Force — Merchant Side (P2)

- Login attempt limits
- OTP request limits
- API rate limiting (per merchant / per IP)
- Suspicious activity alerts
- Temporary block + unlock mechanism

Phase 2 Suggested Order: Webhook Reliability → Smart Routing → Status Matrix → KYC Forward → High TPS QR → Split + Ledger → Schedule Engine → Rate Limiting

PHASE 3 — Professional + Compliance (Month 3–5)

Goal: Reach near Razorpay / Cashfree professional and compliance level. Strong KYC, audit, risk, and advanced product features.

1. DigiLocker + Auto KYC Verification (P2)

- Integrate DigiLocker via partner (Digio / Signzy / Karza)
- Aadhaar, PAN, Driving License pull with consent + OTP
- Auto-fill KYC form with fetched data
- PAN, GSTIN, Udyam, Bank verification APIs
- Name matching logic
- Risk flags (name mismatch, expired docs)
- Optional auto-approve for low-risk merchants
- High-risk cases go to manual review

2. eSign for Merchant Agreement (P2)

- Choose eSign partner (Digio / Leegality)
- Put Merchant Agreement into eSign flow
- Aadhaar eSign or Electronic Signature
- Save signed PDF + audit trail
- Version control — re-sign on agreement changes
- Admin visibility of signed copies

3. Immutable Audit Trail (P2)

- Log every sensitive action (Login, KYC, Payout, Settlement, Role change, Settings)
- Record: Who, What, When, IP, Old value → New value
- Logs cannot be edited or deleted
- Searchable + filterable audit page in Admin
- Export for compliance

4. Full Dispute / Chargeback Workflow (P2)

- Dispute raise UI (Merchant + Customer)
- Evidence upload (invoice, delivery proof, chat)
- Timeline + deadline tracking
- Status flow: Open → Under Review → Won / Lost / Accepted
- Auto deduction / refund on final decision
- Handle gateway chargeback webhooks
- Dispute rate reports per merchant

5. Advanced Reporting & Analytics (P2)

- Financial reports (Gross, Net, Commission, Refunds)
- Settlement reports
- Success rate by Gateway / Payment Method
- Merchant cohort / growth reports
- Tax-ready reports (GST etc.)
- Custom date range + Export (CSV / Excel / PDF)
- Dashboard graphs (daily / weekly / monthly)

6. Sub-merchant / Marketplace Model (P3)

- Platform merchant can create sub-merchants
- Sub-merchant KYC + approval flow
- Commission split rules (Platform → Master → Sub)
- Separate ledger for sub-merchants
- Master merchant sees sub-merchant reports
- Admin sees full hierarchy

7. White-label Checkout Customization (P3)

- Merchant logo, color, brand name on checkout
- Custom domain / subdomain support (optional)
- Thank-you page customization
- Payment method order preference
- Hide/show payment methods per merchant

8. Recurring Payments / Subscriptions (P3)

- Create subscription plans
- Customer mandate (UPI Autopay / e-mandate)
- Auto debit schedule
- Failed payment retry logic
- Pause / Cancel / Upgrade subscription
- Invoice generation for recurring

9. Risk Engine & Velocity Rules (P2/P3)

- Velocity rules (same UPI/card high frequency)
- Sudden high amount spike detection
- New merchant high volume → auto hold
- Blacklist / Whitelist (UPI, VPA, Card BIN)
- Risk score per transaction + per merchant
- Auto actions: Hold / Review / Block

10. Reconciliation Engine (P2/P3)

- Gateway settlement file upload or auto-fetch
- Match system transactions vs gateway report
- Mismatch report (Missing, Extra, Amount difference)
- Auto mark reconciled
- Manual resolve option
- Daily reconciliation summary

11. Grievance Redressal Mechanism (Compliance)

- Customer complaint form (no login required)
- Ticket number + status tracking
- Escalation levels (Merchant → Staff → Admin)
- SLA timers
- RBI-style grievance policy page
- Monthly grievance report

12. Reserve / Rolling Reserve (P3)

- Hold percentage reserve on new merchants
- Reserve release schedule (7 / 15 / 30 days)
- Admin manual adjust reserve
- Clear visibility of held amount to merchant

Phase 3 Suggested Order: Audit Trail → Dispute Workflow → DigiLocker + Auto KYC → eSign → Reporting → Risk Engine → Reconciliation → Grievance → Sub-merchant → White-label + Recurring + Reserve

Summary & Current Position

UniWeb currently has a solid foundation comparable to early-stage Decentro / Cashfree style platforms. Core admin, merchant portal, nodal separation concept, MFA, electronic agreement, and settlement workflow already exist.

Biggest current gaps vs Razorpay / Juspay / Cashfree:

- Automatic Payout (still manual UTR based)
- Smart multi-gateway routing + fallback
- High-throughput QR and multiple Virtual Accounts
- Full risk engine and reconciliation
- Advanced KYC automation and eSign

Recommended immediate focus: Complete Phase 1 (especially Security + Payout API). Once real money moves automatically and securely, the platform becomes commercially usable. Phase 2 and Phase 3 then build scale and professional compliance.

Document Control

Version: 1.0 **|** Date: August 2026 **|** Owner: Uniweb Technologist PVT LTD

This document is for internal planning. Update task checkboxes as work progresses.

UniWeb — Risk Engine Complete Specification

Granular Tasks | Database Design | Score Algorithm | Chargeback Prevention
Architecture | Fraud Algorithms | Hindi Summary

1. Granular Task List (Developer ■■ ■■■)

Phase A — Basic Rules (Week 1–2)

- Create tables: risk_rules, risk_events, risk_blacklist, risk_scores, risk_merchant_limits
- Build velocity checker service (same UPI / phone / IP in time window)
- Implement high-amount threshold rule (configurable per merchant tier)
- New merchant volume hold (first 7–15 days, % hold on settlement)
- Simple blacklist (UPI, phone) — add/remove from Admin
- Log every rule trigger into risk_events with txn_id, rule_id, score, action
- Admin page: list recent risk events + filter by merchant/date/action

Phase B — Score + Auto Actions (Week 3–4)

- Implement risk score calculator (0–100) using weighted factors
- Device fingerprint / IP tracking on checkout + payment
- Auto actions: Allow / Flag / Hold / Block based on score bands
- Hold = capture payment but block settlement until admin release
- Admin rule config UI (edit thresholds without code deploy)
- Manual override API + UI (release hold / unblock) with reason + audit
- Merchant risk score rollup (daily job)

Phase C — Advanced (Month 2+)

- Chargeback feedback loop (lost chargeback → increase merchant risk)
- Pattern detection (same amount repeated, round amounts, etc.)
- Merchant health score integration
- Webhook/alert to Slack/Email on high-risk events
- Export risk reports for compliance

2. Database Fields + Sample Rule Config

Table: risk_rules

```
id | name | rule_type (velocity/amount/merchant/blacklist) | field (upi/phone/ip/amount) | window_seconds |
threshold_count | threshold_amount | score_points | action (allow/flag/hold/block) | is_active | applies_to
(all/new_merchant/tier) | created_at
```

Table: risk_events

```
id | txn_id | merchant_id | rule_id | score | action_taken | factors_json | ip | device_hash | upi_or_phone | created_at |
reviewed_by | reviewed_at | review_notes
```

Table: risk_blacklist

```
id | type (upi/phone/ip/device/card_bin) | value_hash | reason | source (manual/auto_fraud/chargeback) | is_active |
created_by | created_at | expires_at
```

Table: risk_merchant_limits

```
id | merchant_id | daily_volume_limit | per_txn_limit | hold_percent | is_new_merchant | new_until_date | risk_score |
status (normal/limited/frozen)
```

Sample Rule Config (JSON example)

```
{ "name": "same_upi_velocity", "rule_type": "velocity", "field": "upi", "window_seconds": 600, "threshold_count": 5,
"score_points": 20, "action": "hold" }
{ "name": "new_merchant_high_amount", "rule_type": "amount", "threshold_amount": 100000, "applies_to": "new_merchant",
"score_points": 25, "action": "hold" }
{ "name": "blacklist_upi", "rule_type": "blacklist", "field": "upi", "score_points": 50, "action": "block" }
```

3. Risk Score Algorithm Design

Final Score = sum of triggered factor points, capped at 0–100. Negative points (good history) reduce score.

Factor	Condition	Points	Notes
New device / first UPI	First seen device or VPA	+15	Cold start risk
Amount vs avg	Txn > 3x merchant 7-day avg	+10 to +25	Scale by ratio
Velocity breach	Any velocity rule hit	+20	Per rule or max once
Blacklist hit	UPI/phone/IP in blacklist	+50	Almost always block
Night + high amount	00:00–05:00 + above threshold	+10	Optional
New merchant	Account age < 15 days	+10	Combined with amount
Good history	Same customer 5+ success, 0 dispute	–10	Trust discount
Merchant high dispute	Merchant dispute rate > 1%	+15	Merchant-level

Score Band	Action	Settlement Impact
0–30	Allow	Normal settlement
31–60	Flag (visible to ops)	Normal settlement
61–80	Hold	No settlement until release
81–100	Block	Payment declined

Score Band	Action	Settlement Impact
0–30	Allow	Normal settlement
31–60	Flag (visible to ops)	Normal settlement
61–80	Hold	No settlement until release
81–100	Block	Payment declined

Pseudo-code: *score = 0; for each rule if matched: score += points; score = clamp(score, 0, 100); action = band(score); write risk_events; execute action.*

4. Chargeback Prevention Strategy

Prevention (Payment time)

- Clear descriptor / merchant name on customer bank statement
- Strong OTP / 2FA where applicable (UPI already PIN)
- Velocity + amount rules to stop card testing and abuse
- Block high-risk MCC or force extra verification
- Store evidence: IP, device, order details, delivery proof hooks

Early Warning (Post payment)

- Monitor dispute rate per merchant (alert if > 0.5–1%)
- Auto-hold future settlements if dispute rate spikes
- Notify merchant immediately on dispute with evidence checklist
- SLA for merchant to upload invoice / delivery proof

Response (After chargeback raised)

- One-click evidence pack (txn detail + customer info + delivery + communication)
- Track representment deadline
- Lost chargeback → add customer identifier to soft blacklist + increase merchant risk score
- Repeat offender merchant → freeze or raise reserve

5. Risk Engine Architecture (Logical Flow)

```
Checkout / Payment Request
  ↓
Collect signals: amount, UPI/VPA, phone, IP, device_hash, merchant_id, time
  ↓
Risk Engine Service
  ■■ Load active rules + merchant limits + blacklist
  ■■ Run velocity checks (Redis/counter recommended)
  ■■ Run amount + merchant rules
  ■■ Compute score
  ■■ Decide action: Allow / Flag / Hold / Block
  ↓
If Block → decline payment
If Hold → capture but mark settlement_hold=1
If Flag/Allow → normal capture
  ↓
Write risk_events + update merchant risk rollup
  ↓
```

```
 Admin Dashboard / Alerts (email, WhatsApp, Slack)
```

Recommended: keep counters in Redis for velocity; persistent rules/events in MySQL; async alerts via queue.

6. Fraud Detection Algorithms Overview

Algorithm / Method	[redacted]	[redacted]
Rule-based (Velocity)	Fixed limits on count/amount in time window	Phase A — [redacted]
Threshold / Spike	Amount vs historical average	Phase A
Blacklist matching	Exact match on UPI/phone/IP/device	Phase A
Risk scoring (weighted)	Multiple signals → single score → action band	Phase B
Device fingerprinting	Browser/device hash to detect multi-account	Phase B
Graph / link analysis	Same phone linked to many merchants/customers	Phase C
Supervised ML	Train on past fraud/chargeback labels	Later / high volume
Anomaly detection	Unsupervised: unusual pattern vs peer group	Later

[redacted] *Rule-based + Score* [redacted] *ML* [redacted] *labelled fraud data* [redacted] *volume* [redacted]

7. Hindi Summary — [redacted]

[redacted]?
[redacted] — [redacted] UPI [redacted], [redacted], [redacted]

[redacted]:
- [redacted] — [redacted] UPI/[redacted] → [redacted]/[redacted]
- [redacted] — [redacted] → [redacted]/[redacted]
- [redacted] — [redacted] → [redacted]
- [redacted] — [redacted] UPI/[redacted]/IP [redacted]

[redacted] (0–100):
0–30 = [redacted] | 31–60 = [redacted] | 61–80 = [redacted] ([redacted]) | 81–100 = [redacted]

[redacted]: Allow, Flag, Hold, Block, Limit, Freeze Merchant

[redacted]: [redacted] + [redacted] + [redacted] + [redacted]

[redacted]:
[redacted] + [redacted] + [redacted] + [redacted] → [redacted] + [redacted] → [redacted]/ML [redacted]

Document Control

Version: 1.0 | August 2026 | Uniweb Technologist PVT LTD
Companion to: Priority Roadmap, Extra Gaps, Risk Engine Details PDFs

UniWeb — Risk Engine Details

Complete specification for fraud prevention, velocity checks, risk scoring and auto-actions

1. Risk Engine ■■■■ ■■■■ ■■?

Risk Engine ■■■■■■ ■■ ■■■ suspicious activity ■■■■■■ ■■ automatically action ■■■■ ■■ ■■■■■■ ■■■■ ■■ (Hold / Review / Block), ■■■■ fraud, chargeback ■■ loss ■■ ■■■

■■■	■■■■
Velocity Checks	■■ ■■ UPI / Card / Device ■■ ■■■■ ■■■ payment ■■■■■
Amount Spike Detection	■■■■■ ■■■■ ■■■■ amount ■■■
New Merchant Monitoring	■■ merchant ■■ high volume → auto hold
Blacklist / Whitelist	■■■■ UPI, VPA, Card BIN ■■ block ■■■■
Risk Score	■■ transaction + ■■ merchant ■■ 0–100 score ■■■■
Auto Actions	Score ■■ ■■■■■ ■■ Hold / Manual Review / Block

2. Velocity Rules (Speed Limits)

Rule	Example Threshold	Action
Same UPI / VPA	5 payments in 10 minutes	Hold ■■ Block
Same Card BIN	8 payments in 1 hour	Review
Same Device / IP	10 payments in 30 minutes	Hold
Same Customer Phone	6 payments in 1 hour	Review
Same Merchant high count	50 txns in 5 minutes	Soft block + alert

3. Amount Rules

Rule	Example	Action
Single txn too high	> ■1,00,000 (new merchant)	Manual review
Sudden spike	Avg ■5,000 → ■■■■■ ■80,000	Hold
Daily volume spike	Normal 2L → ■■ 15L	Alert + partial hold

4. Merchant Level Rules

Rule	Example	Action
New merchant (first 7–15 days)	Volume > ■2L/day	Auto hold 20–30%
High dispute rate	> 1.5% chargebacks	Limit volume / freeze
KYC incomplete / weak	Documents pending	Only Test Mode
Blacklisted category	Restricted MCC	Block live

5. Blacklist / Whitelist

- Blacklist: UPI ID, VPA, Card number (hashed), Phone, Device ID, IP
- Whitelist: Trusted customers / merchants (rules skip ■■ ■■■■)
- Confirmed fraud ■■ auto-add to blacklist
- Admin manually add/remove ■■ ■■■

6. Risk Score System (0–100)

■■ transaction ■■ score ■■■ Factors:

Factor	Points (Example)

New device / first time UPI	+15
High amount vs merchant average	+10 to +25
Velocity breach	+20
Blacklisted signal	+50 (auto high risk)
Night time + high amount	+10
Good history customer	–10

Score	Action
0–30	Auto Approve
31–60	Soft review / Flag (payment ■■■■ ■■)
61–80	Hold for manual review
81–100	Block + Alert

7. Auto Actions ■■ ■■■■■■ ■■ ■■■■ ■■

1. **Allow** — Normal flow
2. **Flag** — Admin/Staff ■■ ■■■■, payment ■■■■ ■■
3. **Hold** — ■■■■ capture ■■, settlement ■ ■■ ■■ ■■ review ■ ■■
4. **Block** — Payment reject
5. **Limit** — Merchant ■■ daily/monthly limit ■■ ■■ ■■
6. **Freeze Merchant** — Live mode ■■■ (■■■■■ Test Mode)

8. Admin Panel ■■■ ■■■■ ■■■■■ ■■■■■

- Real-time Risk Dashboard (■■ ■■ high-risk transactions)
- Merchant Risk Score list
- Velocity breach alerts
- Blacklist management (Add / Remove)
- Rule configuration UI (threshold ■■■■■ ■■)
- Manual override (Hold ■■■■ / Block ■■■■) + reason
- Audit log (■■■-■■ rule ■■ trigger ■■■)

9. Implementation Priority (Practical)

Phase A (■■■■■ — 1–2 weeks)

- Basic velocity rules (same UPI / same phone)
- High amount threshold
- New merchant volume hold
- Simple blacklist (UPI + Phone)

Phase B (■■■■)

- Full risk score calculation
- Device / IP tracking
- Auto actions (Hold / Block)
- Admin rule configuration UI

Phase C (■■■ ■■■)

- Pattern / ML based detection
- Chargeback prediction
- Merchant health score integration

10. Example Flow

1. Customer ■75,000 ■■ payment ■■■■ ■■
2. System ■■■■■ ■■: Merchant ■■■ ■■ (7 ■■■), ■■ UPI ■■ ■■ 4th payment, Amount average ■■ 8 ■■■■ ■■■■■■
3. Risk Score = 72
4. Action = **Hold**

5. Admin ■■ alert ■■■■ ■■
6. Admin review ■■■■ Allow ■■ Refund ■■■■ ■■

■■■■■■: *Risk Engine = Rules + Score + Auto Action■ ■■■■ ■■■■ Velocity + Amount + New Merchant Hold + Blacklist ■■■■■ Score system ■■ advanced rules ■■■ ■■■ ■■■■ ■■■■ ■■■*

Document Control

Version: 1.0 | Date: August 2026 | Owner: Uniweb Technologist PVT LTD
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